Dornbusch Fischer Startz

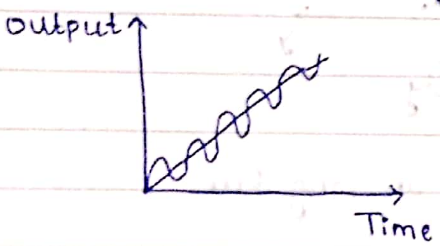
HSS125 - IM

Introductory Macroeconomics :-
↓
Large scale eco study → Measure of o/p

Output :- No 1 single kind → All possible types clubbed together
Growth of Output , Inflation
Unemployment , Exchange Rate , Trade Balance
(Appreciation / Depreciation)

Essentials - Fundamentals of Macroeconomics
Goods Market → Labor Market → Assets Market
Markets of commodities as a whole

Policy Implications



Growth Rate → Long Run

Short Run - Medium Run - Long Run

National Income Accounting

3 Models

- Short Run Model
- Medium Run Model
- Long Run Model (Year / Decade)
- Very Long Run Model (Century / Some centuries) (Growth) → Leads to positive influence on society

Makes a difference in lifestyle of society over a lifetime but not in short terms like 1 or 2

Small Differences in Growth Rate can make a substantial difference in long run.

$N \rightarrow$ Land

$Y = F(L, K, Land)$

O/P — Labor — Capital

To increase o/p we need enhancement of Labor, capital, land

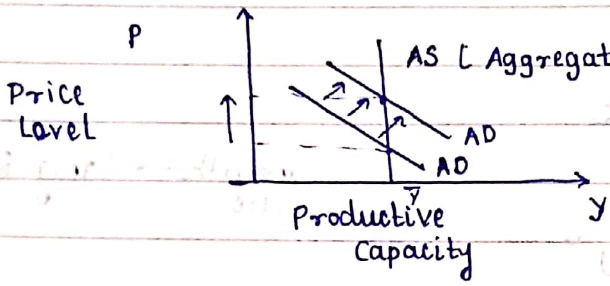
Labor: Grows already over a period of time

1. Capital Accumulation
2. Technological Progress
3. Infrastructure Dev

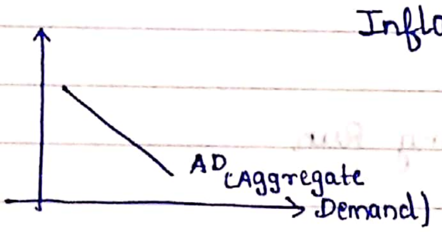
2nd in priority for some nations $\rightarrow$ (3) Infrastructure Dev

Fixed Productive Capacity

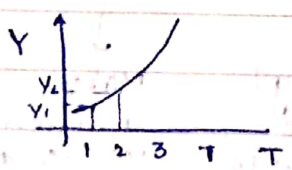
Arises in the Long Run

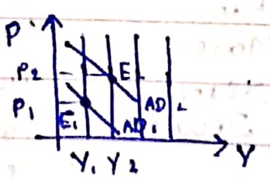


Inflation in Long Run
↓
AD phenomenon
as, price increases as demand increases



In the long run both AS and AD determine prices
In long run AS determines o/p.





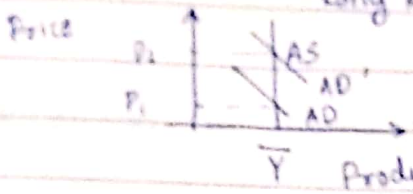
AD → eqbm in goods & money markets

10/01/2024

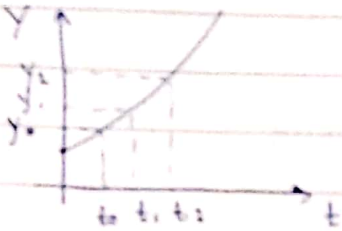
Long Run Growth

Output in long run → constant

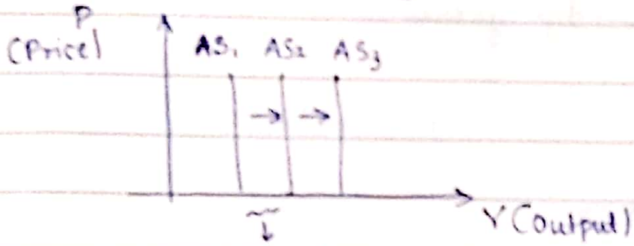
Long Run



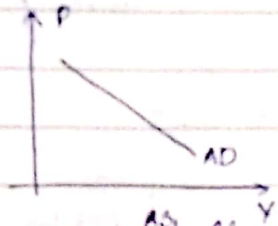
Productive Capacity

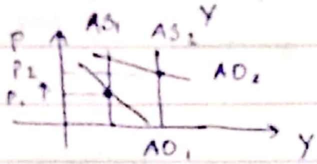


Inflation → Mostly Short Run Phenomenon → AD phenomenon



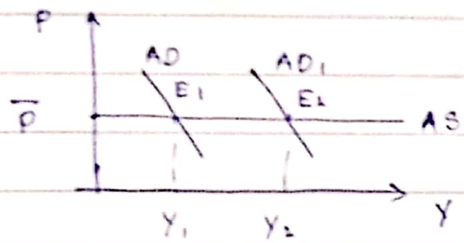
short
2-3% increase





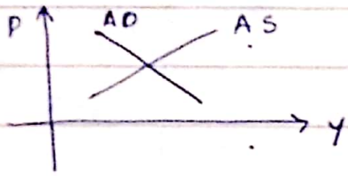
Price increases with demand

Short Run Growth



O/p depends on AD

Medium Run Growth



Price has to increase if O/p cannot increase further in AS

Philip's Curve

11/01/2024

Prices need time to adjust → Hence the corresponding AS curves from SR → MR → LR

Growth (GDP)

↑ in efficiency, productivity

$Y = A\,F(L, K, N)$ — L: Labor, K: capital, N: Land

A ↓ Productivity ↓ through knowledge grows

L↑ ↓ Population increases,

K↑ ↓ Capital increases over time

N↑ ↓ Due to urbanization

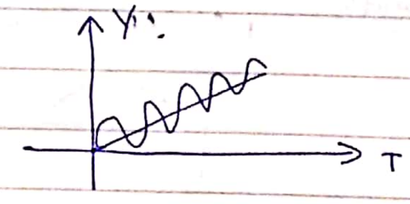
	Brazil	Ghana
1913		
1980		

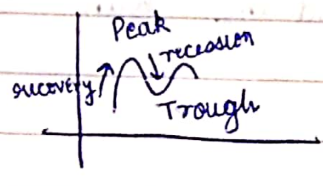
$Y_{B1980} = 5\,Y_{B1913}$

$Y_{G1980} = 1.2\,Y_{G1915}$

Growth
Unemployment
Inflation

Business Cycle
↳ Made of booms and busts till they reach peak





Recovery

Recession → jobs less, unemployment ↑, inflation ↓, drop in growth rate

Output mostly either in peak / trough section

Expansionary Policy

Contractory Policy

12/01/2024

Production Side

inputs → L, K   Land, capital

w, r
↓ wage rate   ↓ rental rate

$Y = f(L, K)$
output ↓

GDP → Gross Domestic Product → value of goods & services products in a particular country

GNP → Gross National Product → Productive units set in other geographical locations are considered as well

$Y = wL + rK + \pi$

GNP = GDP + net property income from abroad

NDP = GDP - depreciation of K

$Y = C + I + G + NX$

C → Consumption, I → Investment, G → Government, NX → Net export

$NX = X - M$

X ↑ Money made by exporting goods abroad

M ↑ value spent on foreign goods

Output ⇄ Spending   Output stimulates spending as it regulates the amt that we spend and spending in turn contributes to o/p

$AD = C + I + G + NX$

$\underline{Y = AD} = C + I + G + NX$

↓

Equilibrium

$Y \neq AD$ Output produced

$Y > AD \rightarrow Y - AD = IU$ (o/p demanded)

IU = Unplanned inventory

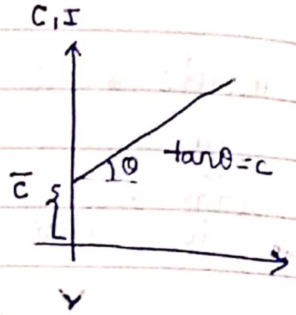
$IU > 0 \rightarrow$ Economy cuts on production

$IU < 0 \rightarrow$ Inventories need to be run down and produce more

Ignoring govt and net export

$C = \bar{C} + c\underline{Y}$ ⇝ amt spent on goods (luxuries)

Consumption ↗ ; $\bar{C}$ ↖ constant component sort of fix payments



c = Marginal propensity to consume — a ratio

$0 < c < 1$

$c = mpc$

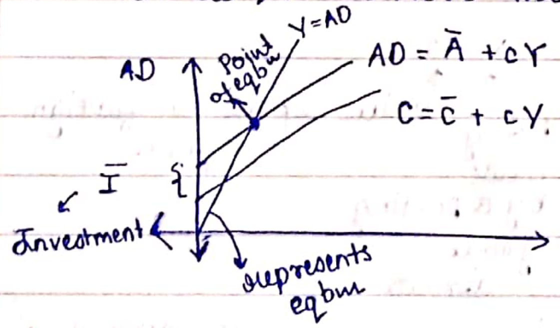
c spent on consumption $1 - c \rightarrow$ saved

$S = Y - C$

$S = Y - \bar{C} - cY$

Saving ↗ $= -\bar{C} + (1 - c)Y$

$S\uparrow$ as $Y\uparrow$ → More income, More saving



$\bar{A} > \bar{C}$

18th → No Class Jan

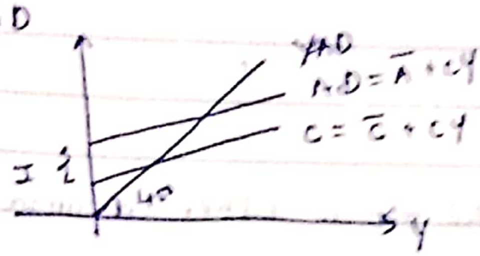
19/01/24

$C = \bar{C} + cY$ → Consumption fn

$0 < c < 1$

$c = mpc$

$AD = \bar{A} + cY$ → AD fn



$AD = C + I$

C ↓ Consumption

I → Investment

Investment $= \bar{I}$

Govt Purchase $= \bar{G}$

Taxes $= \overline{TA}$

Transfer $= \overline{TR}$ ← unemployed people money aid

Net Exports $= \overline{NX}$

Disposable Income → income means to be sent. Subtracting taxes and adding transfer gives Y_D

$Y_D = Y - \overline{TA} + \overline{TR}$

$C = \bar{C} + cY_D$

$= \bar{C} + c(Y - \overline{TA} + \overline{TR})$

$AD = C + I + G + NX$

$= \bar{C} + c(Y - \overline{TA} + \overline{TR}) + \bar{I} + \bar{G} + \overline{NX}$

$= [\bar{C} - c\overline{TA} + c\overline{TR} + \bar{I} + \bar{G} + \overline{NX}] + cY$

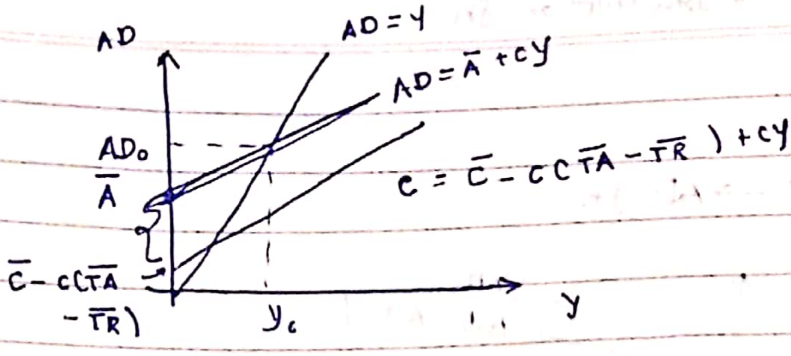
$= [\bar{C} - c(\overline{TA} - \overline{TR}) + \bar{I} + \bar{G} + \overline{NX}] + cY$

$= \bar{A} + cY$

Consumption → Autonomous → $\bar{C}$ → fixed (bound to spend like money on daily necessities)

↳ Income dependent → cY_D

New AD vs Y



$Y = C + I$ All o/p is either consumed or invested

$Y = C + S$

$C + I = Y = C + S$

$I = Y - C = S$

$S \rightarrow$ savings

Some people invest using money of people who save in more sophisticated economy

$Y = C + I + G + NX$

$Y_D = Y + TR - TA$

$Y_D = C + S$

$Y = Y_D - TR + TA$

$= C + S - TR + TA$

$Y = C + I + G + NX = C + S - TR + TA$

$S - I = G + NX + TR - TA$

$= \underbrace{\underbrace{(G + TR}_{\text{Amt leaving govt}} \underbrace{- TA)}_{\text{Amt entering govt}}}_{\text{Budget deficit (BD)}} + NX$

Next Wed No class

$S = \underbrace{I}_{\text{Investments to private investors}} + BD + \underbrace{NX}_{\text{Foreigners loan}}$

10/01/2024

Inflation & Price Indices

GDP - Total value of o/p produced

Real GDP - Give understanding of o/p
Nominal GDP -

q - quantity produced
P - average price
pq - values of o/p   if p↑ pq↑ GDP↑ but in real terms it has not increased much (only price has increased)

* Real GDP - Measures value of all o/ps at a constant level/price then we capture variation in qty only (uses fixed prices)
* Nominal GDP - Calculated for a particular year measured in terms of prices of that particular year → current prices (Market prices)

Inflation - rate of change of avg prices

$$\pi = \frac{\text{current avg price level} - \text{last year avg price level}}{\text{last year avg price level}}$$

last year inflatⁿ

$$= \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$P_t = P_{t-1}[1+\pi]$$ → Inflatⁿ needs to be 0 for constant market price

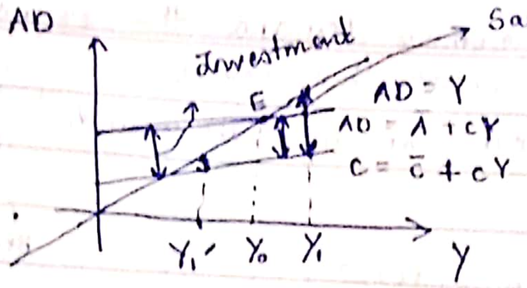
Price Indices
- GDP deflector
- CPI — Consumer Price Index
- PPI - Producer Price Index

} type of measures

GDP deflector - the value of a nominal GDP of a "base yr" ~~~~ normalized by value of real gdp.
↓ On your choice
= Value of a Nominal GDP of base year

01/02/2024

Savings = Income − consumption



Change in eqbm income $\rightarrow \Delta Y_0 = \left\{ \frac{1}{1-c} \right\} \Delta \bar{A}$ Change in

↳ Autonomous expenditure

↓

Can change values as economy changes

$AD - C = I$

↳ Investment

At E $\quad I = S$ → Savings

At Y_1 $\quad S > I$.

Basically for $\forall\, Y_1 > Y_0 \quad S > I$

At Y_1' $\quad I > S$

At Y_0 $\quad S = I$

$Y = C + S + TA$ → Household

- S → Savings
- C → Consumption
- TA → Taxes − TR
- TR → Transfer

$Y = C + I + G$ → Economy level

- G → Government Purchases

$C + I + G = C + S + TA - TR$

$I = S + \underbrace{(TA - TR - G)}_{BS} - NX$

- NX → Net exports
- BS → Budget Surplus

In a simplistic corn economy

S = Corn saved by people

BS = Excess of corn owned by govt

NX = Corn imported

$\Delta A = ₹ 1 \rightarrow$ We spend 1 rupee how much richer does Indian eco become?

c = marginal propensity

$0 < c < 1$

Rounds

↑ in Autonomous Demand this round

↳ ↑ in production this round

↳ Total ↑ in eqbm income (all rounds)

Round | $\Delta \bar{A}$ | $\Delta \bar{A}$ | $\Delta \bar{A}$

Rounds → Increase in Spending

$$\Delta AD = \Delta \bar{A} \left(1 + c + c^2 + \dots\right)$$

$$= \Delta \bar{A} \left(\frac{1}{1-c}\right) = \Delta Y_0$$

↗ Multiplier ↳ Keynesian Multiplier α

↑ in AD → Multiple of initial ↑ in AD

$$\alpha = \frac{1}{1-c}$$

$$\Delta Y_0 = \alpha \Delta \bar{A}$$

$$c\uparrow \Rightarrow \alpha \uparrow$$

02/02/2024

Keynasian Multiplier

$$\alpha = \left[\frac{1}{1-c}\right] \rightarrow \text{Multiplier}$$

		↑ in income
$\Delta \bar{A} \Rightarrow$	$\Delta \bar{A}$	$\Delta \bar{A}$
↓ ↑ in autonomous expenditure		
$c\Delta \bar{A}$	$c\Delta \bar{A}$	$(1+c)\Delta \bar{A}$
$c^2\Delta \bar{A}$	$c^2\Delta \bar{A}$	$[1+c+c^2]\Delta \bar{A}$

$$\Delta AD = \frac{1}{1-c} \Delta \bar{A} = \Delta Y$$

c = Marginal propensity consumed

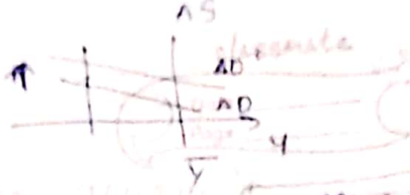
$$\alpha = \frac{1}{1-c}$$

$$c\uparrow \rightarrow \alpha\uparrow$$

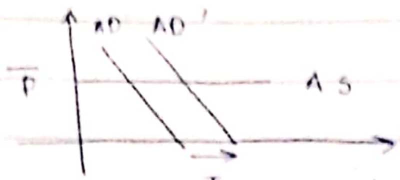
In short run prices are sticky →

Round 3
$c^2 \Delta\bar{A}$
$c^3 \Delta\bar{A}$

$\Delta\bar{A} + c\Delta\bar{A} + c^2\Delta\bar{A}$



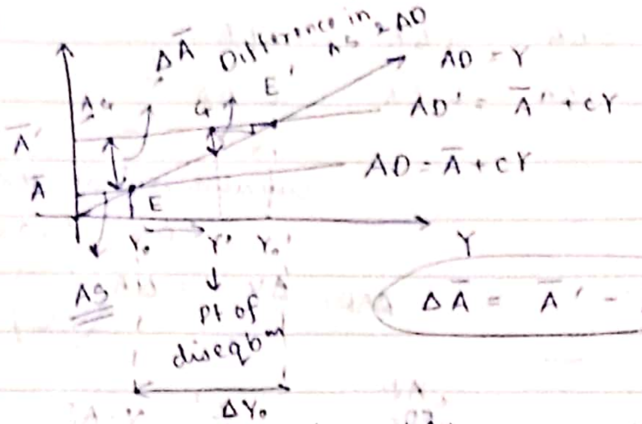
In short run



Increase in o/p on ↑ in AD

Stagflation
↓
Inflation accompanied by unemployment

Multipliers are used when prices are constant



Here only 1 of increase in $\Delta\bar{A}$ is shown

$\Delta\bar{A} = \bar{A}' - \bar{A}$

As we approach eqbm, difference in AS & AD ↓

$\Delta\bar{A}$ ↑ happens only once i.e to increase o/p we need to change or ↑ $\Delta\bar{A}$, only once ∴ As in successive rounds $\Delta\bar{A}$ is multiplied by c i.e autonomous expenditure decreases.

If $\Delta\bar{A}$ had increased every time the gap betⁿ supply and demand would increase every time and eqbm would never be reached

mpc < 1 ⟶ Then a sufficient ↑ in o/p would decrease gap betⁿ AS, AD

$$Y_0 = \frac{\bar{A}}{1 - c(1-t)}$$

07/02/2024

Keynesian Multiplier

$$\frac{1}{1-c}$$

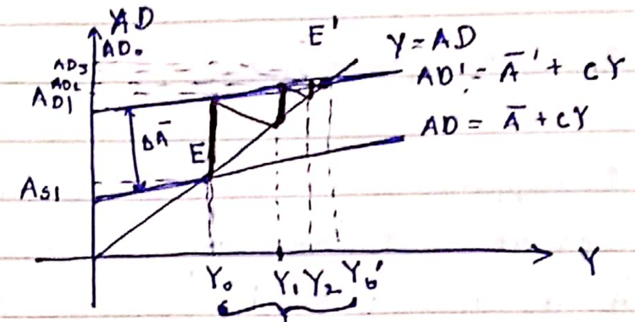
$$\Delta Y = \frac{1}{1-c}\,\Delta\bar{A}$$

	Expenditure	Prodn / OP	Income
R1	$\Delta\bar{A}$	$\Delta\bar{A}$	$\Delta\bar{A}$
R2	$c\Delta\bar{A}$	$c\Delta\bar{A}$	$c\Delta\bar{A}$
R3	$c^2\Delta\bar{A}$	$c^2\Delta\bar{A}$	$c^2\Delta\bar{A}$

Cumulative ↑ in income

$$= \Delta\bar{A}[1 + c + c^2 + c^3 + \dots]$$

$$\Delta AD = \Delta Y = \Delta\bar{A}\left(\frac{1}{1-c}\right)$$



Max^m o/p available from economy's prod cap during that financial yr

$$\Delta Y = Y_0' - Y_0 = \Delta\bar{A}\left[\frac{1}{1-c}\right]$$

$$c\uparrow \rightarrow \Delta Y\uparrow$$

$$\bar{A} = \bar{G} + \bar{I} + \overline{NX}$$

$G = \bar{G}$

$TR = \overline{TR}$

$TA = t \cdot Y$

$$C = \bar{C} + cY_D$$

$$Y_D = Y + TR - TA$$

Consumption fn

$$C = \bar{C} + c[Y + TR - TA)$$

$$C = \bar{C} + c\overline{TR} + c(1-t)Y$$

$cY \rightarrow$ consumed

$Y - cY$

$Y(1-c) \rightarrow$ saved

$tY \rightarrow$ pay taxes

$\downarrow$

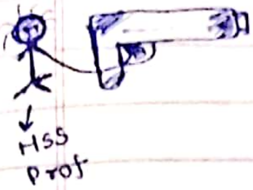
$Y - tY \rightarrow$ amt left to consume after paying taxes

$c(1-t)Y \rightarrow$ amt spent or consumed

$$AD = C + I + G + NX$$

$$= [\bar{C} + c\bar{TR} + c(1-t)Y] + \bar{I} + \bar{G} + \bar{NX}$$

$$= (\bar{C} + \bar{G} + \bar{I} + \bar{NX} + c\bar{TR}) + c(1-t)Y$$



$$Y = \bar{A} + c(1-t)Y$$

as $Y = AD$ ✓

$$Y_0 = \frac{\bar{A}}{1 - c(1-t)}$$

final eqbm income

$$\Delta Y = \Delta \bar{A} \frac{1}{1 - c(1-t)}$$

$t \uparrow$ Multiplier effect $\downarrow$

$\downarrow$

$Y_0 \downarrow$

$\bar{G} \uparrow \rightarrow Y_0 \uparrow$

$TR \uparrow \rightarrow Y_0 \uparrow$

8/02/2024

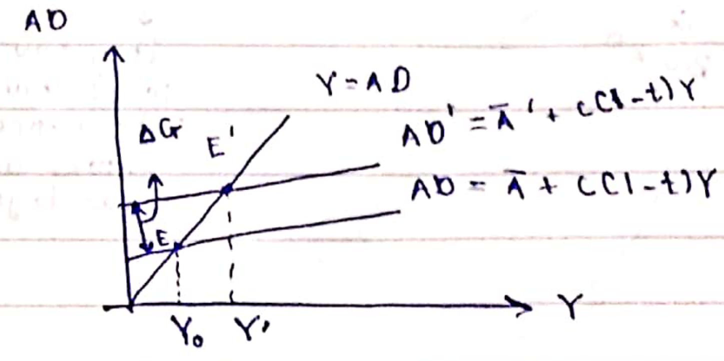
Automatic Stabilizer ⇝ ongoing govt policies that automatically adjust tax rates & transfer payments in a manner that is intended to stabilize incomes, consumption & business spending over the business cycle.

- → Income tax
- → Subsidies & Transfers

Ever since WW II a lot of countries have suffered from recession. Recession is more common, now-a-days than before WW II

$$\Delta Y = \alpha_G \Delta G$$

$$\alpha_G = \frac{1}{1 - c(1-t)}$$



$c = 0.8$, $t = 0.24$ $\rightarrow \alpha_G = 2.5$

$\Delta G = \Delta TR$ → ΔG and ΔTR are taken to be sa[me]

$\Delta AD = c\Delta TR$

$\therefore \Delta Y = \alpha_G (\Delta TR) c$ ∃ Part of TR has to be saved and only the remaining can be transferred to the citizen

$\alpha_G \Delta G = \Delta Y$

$t\uparrow \xrightarrow{\text{direct}} Y_D\downarrow \rightarrow C\downarrow \rightarrow AD\downarrow \rightarrow \Delta Y\downarrow$

indirect ⇒ $\alpha_G\downarrow$

Expansionary Policy
↳ spending is increased by govt although economy is stable like infrastructure dev, education, etc.

9/02/2024

Budget
↳ Primary source of govt income is taxation

$BS = TA - G - TR$

TA: Taxation

G ↓ Government expenditure (Public welfare, development, etc) Corporates can't be trusted on this of course.

TR: Transfer ↓ Sometimes people lose jobs. This is aid for those people like a subsidy.

$BS = tY - G - TR$

tY: Per unit taxation ↓ Taxation based on total eqbm o/p ↓ They have to be channelized into proper sectors and put to good use

Suppose $G\uparrow$ $BS\downarrow$
But as $G\uparrow \rightarrow Y\uparrow$
a scenario may arise like $tY > G$ and $BS > 0$

ΔG = Change in govt expenditure

$\Delta Y_0 = \alpha_G \Delta G$

$\Delta BS = \Delta TA - \Delta G$

as $\Delta G > 0$, $\Delta Y_0 > 0$

so we are earning more and part of earning should go to taxation

$\Delta BS = t\alpha_G \Delta G - \Delta G$
$= (t\alpha_G - 1)\Delta G$

$\alpha = \frac{1}{1 - c(1-t)}$

$$\frac{t}{1-c(1-t)} - 1 = \frac{t-1+c-ct}{1-c(1-t)}$$

$$= \frac{(t-1)(1-c)}{1-c(1-t)}$$

$$= -\frac{(1-t)(1-c)}{1-c(1-t)}$$

$\therefore$ $\underline{\underline{\Delta BS < 0}}$ i.e budget surplus decreases

$c = 0.8$

$t = 0.25$

$\Delta G = Rs\ 1$

$$\Delta BS = -\frac{[1/4][1/5]}{1-4/5[1-3/4]} \times 1$$

$$= \frac{-1/20}{1-4/20}$$

$$= \frac{-1/20}{4/5}$$

$$= -\frac{1}{20} \times \frac{5}{4}$$

$$= \frac{-1}{16} < 1$$

$\Delta BS = \Delta TA^{\uparrow} - \Delta G^{\uparrow}$

s.t $\Delta TA = \Delta G$

$\Delta BS = 0 \rightarrow$ fav

t

$\Delta G = \Delta TA$

$\Delta Y = ?$

$t = 1/\alpha_G$

$\Delta Y = \alpha_G \Delta G$

$t - ct + ct^2 = 1$

$ct^2 - ct + t - 1 = 0$

$t[ct - c + 1] - 1 = 0$

$ct^2 - t(c-1) - 1 = 0$

$$t = \frac{c-1 \pm \sqrt{c^2-2c+1+4c}}{2c}$$

$$= \frac{(c-1) \pm (c+1)}{2c}$$

$= 1$

$\alpha_G = \frac{1}{1-0}$

$= 1$

$\therefore \Delta Y_0 = \Delta G$

$Y_0 \downarrow \Rightarrow C\downarrow$

$AD \downarrow$ by $C\Delta TA$

$= C\Delta G$

O/P ↑ Effect of ↑ in G on Y $= \alpha \Delta G$

O/P ↓ Effect of ↑ in TA on Y $= \alpha c \Delta G$

Net ↑ in O/P $= \alpha \Delta G[1-c]$

$= \alpha \Delta G[1-c]$

$$\boxed{\Delta Y = \Delta G}$$

↓

Here BBM = 1

↗

Balance Budget multiplier

$\Delta TA > 0$

↓

$Y_0 \downarrow \Rightarrow C\downarrow$

$AD\downarrow$ by $C\Delta TA$

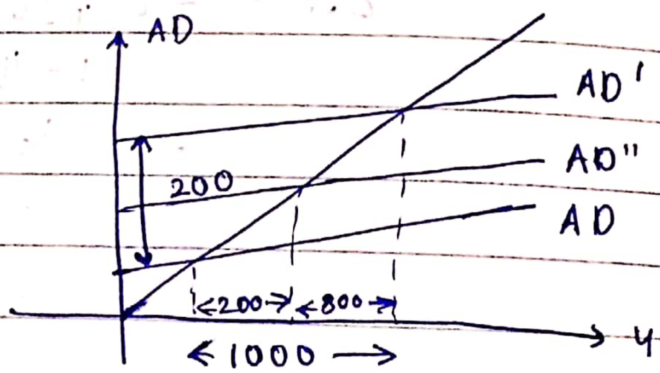
$= C\Delta G$

$C = 0.8$

$\alpha = 5$

$\Delta G = 200$

$\Delta TA = 200$



$\Delta Y = \Delta G$

$= \Delta TA$

$= 200$

$$\boxed{\Delta Y = \underset{A}{\alpha \Delta G} - \alpha c \Delta G}$$